

Fall 2026

Math F113X

Math F113X Exam 1

Name: Solutions

Instructor: _____

Rules:

- Partial credit will be awarded, but you must show your work.
- You may have a $3'' \times 5''$ notecard with writing on both sides.
- Calculators are allowed.
- Turn off anything that might go beep during the exam.

Good luck!

Problem	Possible	Score
1	10	
2	16	
3	20	
4	8	
5	12	
6	15	
7	19	
Extra Credit	(5)	
Total	100	

1. (10 points)

A section of Math F113X with 9 students holds a vote for their favorite vegetable, with the choices being asparagus (A), beets (B) and cabbage (C). The results are as follows:

	Alice	Bob	Cathy	Daniel	Edith	Fahad	Greg	Hyacinth	Ida
1st choice	A	A	A	A	A	B	B	B	C
2nd choice	B	B	C	C	C	A	A	A	A
3rd choice	C	C	B	B	B	C	C	C	B

- a. (4 points) Create a preference schedule with this information.

	2	3	3	1
1 st	A	A	B	C
2 nd	B	C	A	A
3 rd	C	B	C	B

- b. (3 points) How many votes are needed for a majority winner? 5
Provide a brief calculation/explanation for your answer.

9 voters ; $\frac{9}{2} = 4.5$ so 5 votes are needed for a majority

- c. (3 points) What is the smallest number of votes that could give a candidate a plurality? 4
Provide a brief calculation/explanation for your answer.

9 voters & 3 candidates
 $\frac{9}{3} = 3$ so we could have 3 piles of exactly 3 votes and a 3-way tie. So for a plurality you need at least 4.

2. (16 points)

Consider the following preference schedule for an election with 5 candidates and 430 voters.

Rank	120	90	80	67	8	62	3
1st	A	B	C	D	D	E	E
2nd	B	A	B	B	C	B	D
3rd	C	C	A	C	B	D	B
4th	D	D	D	A	A	C	C
5th	E	E	E	E	E	A	A

Answer the following questions. **Show your work.** (This means write down calculations or words so that someone else can understand how you get your answer.)

- a. (4 points) Find the winner using **Plurality voting**. A

Provide a brief calculation/explanation for your answer.

$$A = \underline{120} \quad B = 90 \quad C = 80 \quad D = 67 + 8 = 75, \quad E = 62 + 3 = 65$$

- b. (2 points) How many votes are needed for a **majority**? 216

Provide a brief calculation/explanation for your answer.

$$430/2 = 215 \text{ exactly. Add one voter.}$$

- c. (10 points) Find the winner using **Instant Runoff Voting** (Ranked Choice Voting). B

Clearly show your work for each round and **state who gets eliminated** in each round.

Round 1 (from above)

$$A = 120$$

$$B = 90$$

$$C = 80$$

$$D = 75$$

$$E = 65 \text{ \& eliminated.}$$

Round 2

$$A = 120$$

$$B = 90 + 62 = 152$$

$$C = 80$$

$$D = 75 + 3 = 78$$

D is eliminated

Round 3

$$A = 120$$

$$B = 152 +$$

$$67 + 3$$

$$= 222$$

$$C = 80 + 8$$

$$= 88$$

STOP: B has a majority, since $222 > 215$

3. (20 points)

Consider the following preference schedule for an election with 4 candidates and 24 voters.

points	Rank	6	6	3	9
4	1st	A	B	A	C
3	2nd	B	D	D	A
2	3rd	C	C	C	B
1	4th	D	A	B	D

- a. (6 points) Find the winner using **Borda Count**. C
Clearly show your work.

$$\begin{aligned}
 A &= 6(4) + 6(1) + 3(4) + 9(3) = 69 \quad \text{4-winner!} \\
 B &= 6(3) + 6(4) + 3(1) + 9(2) = 63 \\
 C &= 6(2) + 6(2) + 3(2) + 9(4) = 66 \\
 D &= 6(1) + 6(3) + 3(3) + 9(1) = 42
 \end{aligned}$$

- b. (6 points) Fill in the missing information in the table below and determine the winner using **Copeland's method**.

A vs B	A vs C	A vs D	B vs C	B vs D	C vs D
A: 18 B: 6	A: 9 C: 15	A: 18 D: 6	B: 12 C: 12	B: 21 D: 3	C: 15 D: 9
A	C	A	tie	B	C

$$\begin{array}{l}
 \text{A vs B} \\
 \begin{array}{cccc}
 6 & 6 & 3 & 9 \\
 \hline
 A & B & A & A
 \end{array} \\
 * A = 6 + 3 + 9 = 18 \\
 B = 6
 \end{array}
 \quad
 \begin{array}{l}
 \text{C vs D} \\
 \begin{array}{cccc}
 6 & 6 & 3 & 9 \\
 \hline
 C & D & D & C
 \end{array} \\
 * C = 6 + 9 = 15 \\
 D = 9
 \end{array}$$

Tally:	A:	<u>2</u>
	B:	<u>1.5</u>
	C:	<u>2.5</u>
	D:	<u>0</u>

Winner under Copeland's method: C

- c. (4 points) Is there a **Condorcet winner**? If so, who is it? Briefly explain/justify your answer.

A Condorcet winner is preferred in every head-to-head comparison and wins. No one meets this criterion, because A loses to C, C & B tie (≠ win) and D loses to everyone.

- d. (4 points) Did either Borda Count or Copeland's Method violate the **Condorcet criterion** in this situation? If so, which one(s)? Provide a brief explanation of your answer.

Yes, the Condorcet criterion says the winner of an election should be a Condorcet winner, but there is no Condorcet winner in this situation!

4. (8 points)

Choose one of the following methods: **Plurality**, **IRV/RCV**, **Borda Count**, and **Copeland's Method**.

State at least one **strength** your chosen method has over the other three options. State at least one **weakness** your chosen method has.

You should refer to some of the various **fairness conditions** in discussing the weakness of your method, and explain what the fairness condition you are referring to means in the context of your method.

Chosen method: plurality

Strength (4 points):

easy to implement/
familiar

Weakness (4 points):

- winner need not be preferred by a majority
- fails IIA

5. (12 points)

In a certain corporation, shareholders receive 1 vote for each share of stock they hold. A small corporation has two people who have 30 shares each, two people who have 20 shares each, and one person who has only 10 shares.

- a. (3 points) If any decisions require a majority vote, set up a weighted voting system of the form $[q : w_1, \dots, w_n]$ to represent this corporation's shareholder votes.

$$[56 : 30, 30, 20, 20, 10] \quad \begin{array}{l} \text{total weight} = 30 + 30 + 20 + 20 + 10 \\ = 110 \end{array} \quad \begin{array}{l} \text{majority} = \frac{110}{2} + 1 = 56 \end{array}$$

- b. (3 points) Certain decisions, like amending the bylaws, require a $\frac{2}{3}$ vote. How would you change your weighted voting system to represent this situation?

Change quota: $\frac{2}{3}(110) = 73\frac{1}{3}$ so set quota to 74

- c. (3 points) What is the largest value the quota could be for this system? 110

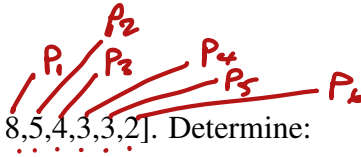
total weight

- d. (3 points) What is the smallest value the quota could be for this system? 56

majority

6. (15 points)

Consider the weighted voting system [18: 8, 5, 4, 3, 3, 2]. Determine:

a. (1 point) The total number of players: 6b. (1 point) The total weight of the system: 25

$$8 + 5 + 4 + 3 + 3 + 2 = 25$$

c. (1 point) The quota: 18

d. (3 points) Are there any dictators? Explain how you know, with a few words.

No, no one can meet quota alone

e. (3 points) The coalition $\{P_1, P_2, P_3, P_6\}$ is a winning coalition. Explain how you know it is a winning coalition using a computation and some words.

$$\text{weight of } P_1, P_2, P_3, P_6 = 8 + 5 + 4 + 2 = 19$$

$19 > 18$ so the coalition is winning

f. (3 points) In the coalition $\{P_1, P_2, P_3, P_6\}$, is P_6 critical? Explain why or why not, using a computation and some words.

Yes, if you remove P_6 , the remaining weight is $17 < 18$
so they no longer meet quota

g. (3 points) Does player P_2 have veto power? Why or why not?

No, $25 - 5 = 20 > 18 = \text{quota}$, so the remaining voters can make a winning coalition without P_2

7. (19 points)

Consider the weighted voting system [27 : 19, 18, 8, 2] and the list of all possible coalitions.

coalition	weight	winning?	coalition	weight	winning?
<u>P₁</u> <u>P₂</u>	37	✓	<u>P₁</u> P ₂ P ₃	45	✓
<u>P₁</u> <u>P₃</u>	27	✓	<u>P₁</u> <u>P₂</u> P ₄	39	✓
P ₁ P ₄	21	✗	<u>P₁</u> <u>P₃</u> P ₄	29	✓
P ₂ P ₃	26	✗	<u>P₂</u> <u>P₃</u> <u>P₄</u>	28	✓
P ₂ P ₄	20	✗	P ₁ P ₂ P ₃ P ₄	47	✓
P ₃ P ₄	10	✗			

- a. (2 points) Put a check ✓ in the column labeled “winning?” to identify all winning coalitions, and mark ✗ for non-winning coalitions.
- b. (6 points) In each winning coalition, **underline the critical players**.
- c. (5 points) Using this information, compute the **Banzhaf Power Index** of each player by filling in the following table.

Player	# critical	power
P ₁	5	5/12 = 42%
P ₂	3	3/12 = 25%
P ₃	3	3/12 = 25%
P ₄	1	1/12 = 8.3%
	<u>12</u>	

- d. (2 points) Is there a **dummy**? Why or why not?

No, everyone is critical in at least one coalition

- e. (4 points) Players 1 and 2 have almost the same number of votes, and Player 3 has less than half as many votes as Player 2. Does their decision-making power reflect this vote distribution? Why or why not/how can you tell?

No: P₃ & P₂ have the same power even though they have different votes, and P₁ has almost twice as much power as P₂ even though their weights only differ by one!

Extra Credit (5 points)

A professional basketball team has five coaches: a head coach (H) and four assistant coaches (A1, A2, A3, and A4).

Constraint: Player personnel decisions require at least three Yes votes, one of which must be from the head coach (H).

If h is the weight assigned to the head coach H , and a is the weight assigned to each of the four assistant coaches (assuming they each get the same number of votes), determine values

$$[q : h, a, a, a, a]$$

that describe the weighted voting system. Explain why your assignment of values matches the constraint given above.

$$[7 : 5, 1, 1, 1, 1]$$

Need H to have veto power!
and need quota to be
at least $h + 2$ so that you need 2 A.
Veto power means need $h > 4$.

Check:

$$HAA = 5 + 1 + 1 = 7 \quad \checkmark$$

$$HA = 6 < 7$$

$$AAAA = 4 < 7$$

so this meets the requirements.